

Identifying a relaxing internal state in free-fall timing: finite-frequency boundaries and an application to PSR J0337+1715

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Abstract

Proven. We formulate finite-frequency identifiability boundaries for a relaxing internal state in nuisance-projected free-fall timing. At K distinct positive carrier frequencies, a nonzero single relaxation pole admits an exact real derivative comparator of degree at most N precisely when $N \geq 2K - 1$. Positivity with a known rate gap restricts this ambiguity. For a positive reciprocal spectrum, two calibrated complex samples test for one observable relaxation time through a moment-variance equality. Finite precision cannot resolve arbitrarily close poles, and periodic observations do not determine transients. These results specialize classical interpolation and moment theory to timing measurements. **Counterexample candidate.** A damped scalar-charge model supplies reciprocal forces and a controlled one-pole reduction. **Proven.** Its leading drive in PSR J0337+1715 satisfies a phase closure violated by our earlier auxiliary analysis. **Imported from prior work.** Auditing that analysis exposes strong dependence on the nuisance span. A calibrated six-coefficient confidence region gives 95.17–95.43 percent inclusion in independent simulations within four specified covariance conditions. With archived timing derivatives, the recorded carrier coefficients exceed its null threshold, but an instantaneous plus single-relaxation response to the leading physical drive fails at all six evaluated lags. Rejection is marginal at 2 days and robust for the sign required by the scalar-charge model. No relaxation is detected, and the excess remains unattributed. **Conjectural.** Derivative uncertainty and incomplete physical matching keep this conditional application from providing an empirical strong-equivalence-principle bound.

Keywords: tests of gravity; strong equivalence principle; scalar-tensor gravity; dynamical identifiability; pulsar timing

1 Introduction

Imported from prior work. Compact-body structure enters worldline effective field theory (EFT) through body-dependent coefficients and, when needed, explicit internal degrees of freedom [1, 2]. Internal modes appear as poles of response functions [3, 4], and nonadiabatic dynamical scalarization supplies a monopolar example [5]. Tests of the strong equivalence principle (SEP) are reviewed by Will [6]. The pulsar triple PSR J0337+1715 has provided sensitive tests of a static SEP parameter [7, 8]; a later planet/noise analysis and public data release provide the setting for the stored dynamic-template calculation used here [9, 10].

Proven. A static description with finitely many operators at a fixed order does not make a relaxing internal state unobservable, and a pole in a body’s response does not by itself make that

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state identifiable from timing. A timing measurement samples the response at a few orbital carriers, after the nuisance directions of the timing model are projected out. Identifiability then depends on the number of carriers, the comparator class admitted for the instantaneous response, the nuisance span and the precision.

Counterexample candidate. We use a one-state relaxation as the benchmark and a damped scalar charge as its physical realization.

Proven. For this benchmark we give an exact boundary for real derivative comparators that interpolate a single pole at K carriers (Section 3.1), a comparator restriction from positivity and a rate gap (Section 3.2), a two-sample test and recovery of one observable relaxation time with its finite-precision limit (Section 3.3), the insufficiency of a finite periodic record for transients (Section 3.4), and a phase gate on the physical drive in J0337 (Section 4). These boundaries specialize classical results, recalled below; what is new is their formulation for a nuisance-projected free-fall timing measurement, the comparator classes fixed by conservative or reciprocal dissipative physics, the drive phase gate, and their application to a stored J0337 analysis. They do not establish a new theory of gravity or a general unobservability theorem.

Imported from prior work. Theorem 1 is real-coefficient polynomial interpolation at conjugate nodes, a special case of rational interpolation and realization [11], and it matches the count by which K sinusoids are persistently exciting of order $2K$ in system identification [12]. The moment-variance inequality is the positivity of a 2×2 Hankel matrix in the truncated Stieltjes moment problem, whose singular case forces a single atom [13]. The close-pole limit is the ill-conditioning familiar from relaxation-spectrum inversion [14], and the transient obstruction is a statement of insufficient excitation.

Imported from prior work. We apply them to our earlier stored J0337 analysis, which is unpublished and recorded in the repository, exposing the dependence of its interval on the nuisance span and calibrating its inference (Section 5). The Supplementary Information (Online Resource 1, abbreviated SM) contains the complete technical account: a finite static operator catalog, the full audits and their provenance, and a worked calculation for the inner white dwarf of the triple. The SM keeps its own section, equation, table and figure numbering.

Each scientific paragraph or result carries a status label. **Proven** denotes conditional mathematics, including numerical evaluation of explicit algebraic identities; **Imported from prior work** denotes literature or numerical experiments recorded in the repository, including this paper’s unpublished audits; **Counterexample candidate** denotes a proposed physical/model realization of a state that a static description would miss; **Conjectural** denotes an interpretation requiring additional work. A theorem’s premises are hypotheses of its conditional statement, not independently established facts.

2 Benchmark and comparators

Counterexample candidate. Let a known dimensionless drive F excite a dimensionless state and a scalar readout q :

$$\tau_\chi \dot{\chi} + \chi = \alpha F(t), \quad q(t) = c_Y F(t) + c_\chi \chi(t), \quad \beta = \alpha c_\chi, \quad \tau_\chi > 0, \quad (1)$$

with settled transfer function

$$G(i\omega) = c_Y + \frac{\beta}{1 + i\omega\tau_\chi}. \quad (2)$$

The readout may correct a specified observable coupling; assigning it to a mass or to a pairwise gravitational parameter are different modeling choices. Relaxation presupposes an effective dissipative setting. The general solution also contains $\chi_h e^{-(t-t_0)/\tau_\chi}$; the periodic analysis assumes that this transient has decayed or is absent.

Proven. A degree- N derivative comparator is a real polynomial $P_N(i\omega)$, the transfer function of a local readout $\sum_{n \leq N} a_n F^{(n)}$ with real a_n . Expanding Equation (2) in $i\omega\tau_\chi$ gives such comparators with error at most $|\beta|\rho^{N+1}$ on $|\omega\tau_\chi| \leq \rho < 1$, so in the adiabatic regime a relaxation is absorbed by a derivative comparator. The results below concern finitely many carriers at arbitrary $\omega\tau_\chi$.

3 Finite-frequency identifiability

3.1 Exact interpolation boundary

Proven. Theorem 1 (real finite-frequency boundary). Suppose c_Y, β are real, $\beta \neq 0$, $\tau_\chi > 0$, and $\omega_1, \dots, \omega_K$ are distinct positive frequencies at which the drive is known and nonzero, so that the values $G(i\omega_k)$ are available. A polynomial $P_N(z)$ of degree at most N with freely chosen shared real coefficients matches $G(i\omega_k)$ at all K frequencies if and only if $N \geq 2K - 1$.

Proven. Define $R(z) = (1 + \tau_\chi z)[P_N(z) - c_Y] - \beta$. Real coefficients imply that a root at $i\omega_k$ is accompanied by $-i\omega_k$. Matching thus gives $2K$ distinct roots, but R has degree at most $N+1$. If $N < 2K - 1$, R would be identically zero, contradicting $R(-1/\tau_\chi) = -\beta \neq 0$. Conversely, set

$$Q_K(z) = \prod_{k=1}^K (z^2 + \omega_k^2), \quad P_{2K-1}(z) = c_Y + \frac{\beta[1 - Q_K(z)/Q_K(-1/\tau_\chi)]}{1 + \tau_\chi z}. \quad (3)$$

The numerator vanishes at $-1/\tau_\chi$, giving a real polynomial of degree $2K-1$ upon division. At every carrier it equals G . The construction allows unrestricted coefficients; power-counting bounds, passivity or external calibration would define a smaller comparator class.

Proven. The first exact obstruction to a real degree- N comparator is $K = \lfloor (N+1)/2 \rfloor + 1$. Three positive carriers exclude N up to four, but a fifth-degree real comparator interpolates them. With complex coefficients and only positive-frequency samples, interpolation permits K up to $N+1$. No finite polynomial equals a nonzero single-pole response on an open frequency interval. Exact noninterpolation does not guarantee detectability: nearly coincident carriers or small $\omega\tau_\chi$ can make the difference arbitrarily small, so precision and nuisance geometry are additional requirements.

3.2 A comparator restriction from positivity and a rate gap

Proven. A real local conservative quadratic action $\frac{1}{2} \int FP(D)F dt$, with $D = d/dt$ and vanishing boundary variations, contributes the self-adjoint response $[P(D) + P(-D)]/2$, so odd derivatives cancel. An even comparator is therefore justified for this time-reversal-invariant conjugate response. Dissipation, nonconjugate readout, a time-dependent background or an unknown drive phase can invalidate this restriction.

Counterexample candidate. A different comparator class follows from positive quadratic state energy, reciprocal forcing/readout and positive Rayleigh dissipation:

$$\mathbf{\Gamma} \dot{\mathbf{q}} + \mathbf{K} \mathbf{q} = \mathbf{b} F, \quad q_{\text{out}} = \mathbf{b}^T \mathbf{q} + c_0 F. \quad (4)$$

Both matrices are real symmetric positive definite. Impose a lower bound Λ on every eigenvalue of $\mathbf{\Gamma}^{-1/2} \mathbf{K} \mathbf{\Gamma}^{-1/2}$. This describes modes that relax faster than a specified scale; the gap is a hypothesis requiring independent physical matching.

Proven. Orthogonal diagonalization gives

$$H(z) = c_0 + \sum_j \frac{a_j}{1 + z\tau_j}, \quad a_j \geq 0, \quad 0 < \tau_j \leq \Lambda^{-1}. \quad (5)$$

For $\omega_h > \omega_l > 0$, put $S(\omega) = -\text{Im } H(i\omega)/\omega$. Then

$$S(\omega_l) \leq R_\Lambda S(\omega_h), \quad R_\Lambda = \frac{1 + (\omega_h/\Lambda)^2}{1 + (\omega_l/\Lambda)^2}. \quad (6)$$

Indeed $(1 + \omega_h^2 \tau^2)/(1 + \omega_l^2 \tau^2)$ increases with positive τ ; summing its bound with the nonnegative weights $a_j \tau_j/(1 + \omega_h^2 \tau_j^2)$ proves the inequality. A positive single pole with $\tau_\chi > \Lambda^{-1}$ violates it, and the real instantaneous coefficient cancels from S . The argument also applies to a convergent nonnegative distribution of relaxation times with the same support.

Proven. This ties the comparator to energy, reciprocity and a rate gap rather than to an arbitrary Taylor cutoff. Negative residues, a different drive/readout pair, oscillatory modes or removal of the gap can evade it. Rejecting this class would not identify one internal state: multiple slow states or a slow memory distribution can also violate the inequality. Neither the rate gap nor reciprocal readout has been empirically established for J0337 here.

3.3 One observable pole from two calibrated samples

Proven. Suppose $H(i\omega) = c_0 + \int a(d\tau)/(1 + i\omega\tau)$, with real c_0 and a nonnegative finite measure on $\tau > 0$. At two calibrated frequencies $0 < l < h$, let $S_\omega = -\text{Im } H(i\omega)/\omega$ and define

$$R = \frac{h^2 S_h - l^2 S_l}{h^2 - l^2}, \quad D = \frac{\text{Re } H(il) - \text{Re } H(ih)}{h^2 - l^2}, \quad Q = \frac{S_l - S_h}{h^2 - l^2}. \quad (7)$$

These are the zeroth, first and second moments of $d\mu = \tau a(d\tau)/[(1 + l^2 \tau^2)(1 + h^2 \tau^2)]$. Therefore

$$RQ - D^2 = R^2 \text{Var}_{\mu/R}(\tau) \geq 0. \quad (8)$$

For $R > 0$, equality holds exactly for one observable relaxation time $\tau_0 = D/R$. Its amplitude is $a_0 = R(1 + l^2 \tau_0^2)(1 + h^2 \tau_0^2)/\tau_0$, and either real part fixes c_0 . A relaxation-time atom at $\tau = 0$ would be indistinguishable from c_0 . This assumes exact common drive/readout calibration; it is not an observed equality in J0337.

Proven. One observable pole does not count hidden, unexcited or degenerate internal variables. Two positive poles at $\tau_0 \pm \epsilon$, each of weight $a_0/2$, differ from one pole by

$$\frac{a_0}{2} \left[\frac{1}{1 + s(\tau_0 - \epsilon)} + \frac{1}{1 + s(\tau_0 + \epsilon)} \right] - \frac{a_0}{1 + s\tau_0} = \frac{a_0 s^2 \epsilon^2}{(1 + s\tau_0)[(1 + s\tau_0)^2 - s^2 \epsilon^2]}. \quad (9)$$

For $0 < \epsilon < \tau_0$ both modes are stable, and the difference vanishes quadratically. A finite-noise test cannot uniformly distinguish exactly one pole from arbitrarily close two-pole alternatives without a separation or weight condition. Without positivity, the stable strictly proper addition $\eta \prod_k (s^2 + \omega_k^2)/(s + \lambda)^{2K+1}$, $\lambda > 0$, vanishes at every carrier but changes the response elsewhere; it need not be a positive reciprocal spectrum.

3.4 Transients outside a finite periodic record

Proven. A homogeneous amplitude decays to fraction ϵ only after $\tau_\chi \log(1/\epsilon)$; without an initial-amplitude bound this gives no uniform absolute signal bound. More fundamentally, the causal differential operator

$$\mathcal{A}(D) = D \prod_k (D^2 + \omega_k^2) \quad (10)$$

annihilates the constant and all carrier inputs, while its action on e^{-t/τ_χ} is nonzero. Linear operators L and $L + \eta \mathcal{A}(D)$ can agree on every carrier and static response and differ arbitrarily on a transient. The stable, strictly proper filter $\eta s \prod_k (s^2 + \omega_k^2)/(s + \lambda)^{2K+2}$, $\lambda > 0$, likewise vanishes at zero frequency and at every carrier but responds to an exponential input, so the conclusion also holds within stable causal systems. A finite periodic response record is therefore insufficient. This does not assert that every such operator is a realization of the timing theory; a validated forward model or a dedicated transient response is needed.

4 A damped scalar-charge realization and its physical drive

Imported from prior work. Compact bodies in scalar-tensor gravity carry scalar charges [15]. Dynamical scalar-charge worldline models add a charge potential and mode inertia; compact-body matching and monopole radiation damping have been developed by Khalil et al. [5]. We use that framework for the following restricted reduction.

Counterexample candidate. In units $G=c=1$, hold the companion charges fixed, retain leading constant orbital masses, and let only the pulsar charge Q_p evolve:

$$L = \sum_A \frac{m_A v_A^2}{2} + \frac{I \dot{Q}_p^2}{2} - V(Q_p) + \sum_{A < B} \frac{m_A m_B + Q_A Q_B}{r_{AB}}, \quad \mathcal{R} = \frac{\Gamma \dot{Q}_p^2}{2}, \quad (11)$$

with positive I and Γ . A constant scalar background is included in V , and the Rayleigh function $\mathcal{R}$ specifies dissipation. Higher post-Newtonian and radiation terms of a complete timing theory are omitted.

Proven. Variation at independent positions and charge gives reciprocal pair forces with $\Delta_{pj} = Q_p Q_j / (m_p m_j)$ and $I \ddot{Q}_p + \Gamma \dot{Q}_p + V'(Q_p) = \sum_j Q_j / r_{pj}$, with $\dot{E}_{\text{orbit+state}} = -\Gamma \dot{Q}_p^2$. Linearizing about a stable equilibrium Q_0 with $\kappa = V''(Q_0) > 0$ gives the transfer $1/(\kappa - I\omega^2 + i\Gamma\omega)$, valid for small force ratios such as $|V'''(Q_0)\delta Q|/(2\kappa) \ll 1$ once homogeneous modes have decayed (SM Section 4.3). It differs from $1/(\kappa + i\Gamma\omega)$ by a relative amount of at most $\epsilon_I/(1 - \epsilon_I)$ when $\epsilon_I = I\omega^2/|\kappa + i\Gamma\omega| < 1$. The bound must be small at every retained carrier, and separated overdamped roots require $I\kappa/\Gamma^2 \ll 1$. If both companion charge/mass ratios equal a_w and $\delta U = \sum_j m_j [1/r_{pj} - \langle 1/r_{pj} \rangle]$, the common pulsar-pair response is

$$\delta \Delta(\omega) = \frac{B \delta U(\omega)}{1 + i\omega \tau_\chi}, \quad B = \frac{a_w^2}{\kappa m_p}, \quad \tau_\chi = \frac{\Gamma}{\kappa}, \quad (12)$$

so that $F = \delta U/U_*$, with the normalization U_* of Equation (13), gives $\beta = BU_* \geq 0$; the signed- β fits of Section 5 admit a larger model. The minimal realization has $c_Y = 0$; co-fitting a free instantaneous coefficient admits a fast response or a broader comparator. If $a_i \neq a_o$, the pair responses differ by $(a_i - a_o)\delta Q_p/m_p$, obstructing a common nonzero modulation. A responsive companion with susceptibility C_j shifts the stiffness by $-\sum_j C_j/r_{pj}^2$, which must be small relative to κ .

Proven. In this class, fixing the potential, equilibrium charge and drive coupling fixes the static susceptibility $1/\kappa$, but $\Gamma = \kappa\tau_0$ with any $\tau_0 > 0$ leaves that equilibrium unchanged. Equilibrium information alone therefore does not establish the rate gap of Section 3.2; a specified microscopic theory may relate these parameters. **Imported from prior work.** The leading coupled monopole damping force of Khalil et al. is proportional to the negative sum of charge velocities [5]. **Proven.** Its two-charge all-ones damping matrix has a null direction, so the positive-definite comparator of Section 3.2 is not automatic.

Proven. Let $\mathbf{r} = \mathbf{x}_p - \mathbf{x}_i$, $\mathbf{R} = \mathbf{x}_b - \mathbf{x}_o$, b be the inner center of mass, and $f = m_i/(m_p + m_i)$. In the aligned coplanar limit, retain terms linear separately in eccentricity and $fa_{\text{in}}/a_{\text{out}}$, dropping their products. With mean longitudes $\lambda_p = n_{\text{in}}(t - t_{\text{asc},p})$, $\lambda_b = n_{\text{out}}(t - t_{\text{asc},b})$ and $M_A = \lambda_A - \varpi_A$,

$$\begin{aligned} \delta U &= A_{\text{in}} \cos M_p + A_{\text{out}} \cos M_b - A_{\text{dif}} \cos(\lambda_p - \lambda_b) + \cdots, \\ A_{\text{in}} &= \frac{m_i e_{\text{in}}}{a_{\text{in}}}, \quad A_{\text{out}} = \frac{m_o e_{\text{out}}}{a_{\text{out}}}, \quad A_{\text{dif}} = \frac{m_o f a_{\text{in}}}{a_{\text{out}}^2}, \end{aligned} \quad (13)$$

and we normalize the drive by $U_* = A_{\text{in}} + A_{\text{out}} + A_{\text{dif}}$. For positive carrier amplitudes the phase closure

$$\mathcal{C} = \phi_{\text{dif}} - \phi_{\text{in}} + \phi_{\text{out}} = \pi + \varpi_p - \varpi_b \pmod{2\pi} \quad (14)$$

is invariant under a common origin shift because $n_{\text{dif}} = n_{\text{in}} - n_{\text{out}}$.

Imported from prior work. In the released timing code the orbital parameters e and κ are $e \sin \varpi$ and $e \cos \varpi$, the ELL1 convention [10, 16], and $t_{\text{asc}} = t_{\text{pericenter}} - P\varpi/(2\pi)$ [9]; the published pericenters are 97.62 and 95.62 degrees [17]. **Proven.** The frozen parameters then give $\varpi_p = 1.69406454$ and $\varpi_b = 1.67084424$ radians (97.06 and 95.73 degrees) and $\mathcal{C} = 3.16481295$ radians. The drive dictionary of our earlier stored analysis instead has $\mathcal{C} = 0$: it took $\varpi = \pi/2$ and the opposite sign of the difference term, so its inner, outer and difference phases differ from Equation (13) by 0.12326821, 0.10004792 and π radians. No common time shift repairs the mismatch. The unit-drive family of Section 5 also has zero closure; its envelope does not contain this leading physical family.

Counterexample candidate. Coupling limits that our earlier analysis derived from that dictionary are therefore withdrawn as constraints on this realization. Rescaling the unit-drive interval cannot replace them; a corrected physical analysis must prescribe the unequal amplitudes, phases, neglected harmonics and additional forces, then validate its inference (Section 5.5).

5 Application to a stored PSR J0337+1715 analysis

5.1 Data, templates and intervals

Imported from prior work. Our earlier stored analysis uses 12,474 public Nançay pulse times spanning approximately 2987.9 days in 2013–2021, a published planet-model baseline, and the released Nutimo implementation [9, 10]. **Counterexample candidate.** It implements the benchmark as a time-dependent pulsar-companion pair coupling, $V_{pj} = -Gm_p m_j [1 + \Delta_0 + q(t)]/r_{pj}$, with fixed inertial masses (SM Section 4.2). **Imported from prior work.** Six stored response columns are central finite differences of the modified integrator: cosine and plus- $\pi/2$ drives at $\omega_{\text{in}} = 3.856137$ rad/day, $\omega_{\text{out}} = 0.019200$ rad/day and $\omega_{\text{dif}} = \omega_{\text{in}} - \omega_{\text{out}}$. They are numerical derivatives of a non-linear model. The two high-frequency carriers differ only by ω_{out} , so the three distinct samples are not three equally informative frequency measurements.

Proven. The plus- $\pi/2$ column $C_{k,s}$ responds to $-\sin \omega_k t$. With $g_{1k} = 1/(1 + \omega_k^2 \tau_\chi^2)$, $g_{2k} = \omega_k \tau_\chi / (1 + \omega_k^2 \tau_\chi^2)$, drive amplitude d_k and phase ϕ_k , the relaxation template is

$$T_\beta = \sum_k d_k [(g_{1k} \cos \phi_k + g_{2k} \sin \phi_k) C_{k,c} + (g_{1k} \sin \phi_k - g_{2k} \cos \phi_k) C_{k,s}], \quad (15)$$

and the co-fitted instantaneous column is $T_{c_Y} = \sum_k d_k (\cos \phi_k C_{k,c} + \sin \phi_k C_{k,s})$.

Imported from prior work. The nuisance block includes 28 timing-parameter derivative columns, an offset, 30 low-frequency sine/cosine pairs, and a static-SEP guard column. A relative singular-value cut of 10^{-3} yields rank 71; the full construction retains all 90 singular vectors, and the 19 discarded directions have not been established to be physically forbidden. For a fixed Gaussian covariance and a flat amplitude prior, the interval half-width U solves

$$\Phi\left(\frac{U - \hat{\beta}}{K\sigma_F}\right) - \Phi\left(\frac{-U - \hat{\beta}}{K\sigma_F}\right) = 0.95, \quad U \geq 0, \quad (16)$$

where Φ is the standard normal distribution function and $\hat{\beta}$ and σ_F come from the two-column fit $[T_{c_Y}, T_\beta]$ after nuisance projection. $K=1$ uses the Fisher width; $K=10$ is a chosen safety factor, not an inferred noise parameter. The scan covers 65 lags from 2 to 500 days and 4821 origins spanning one outer period, with $d_k = 1$ and $\phi_k = \omega_k t_{\text{off}}$; that period is an analysis domain, not a common period of the carriers. The scan records the maximum U on this finite grid. We call it a *registered-grid envelope*; it is not a phase marginalization. The coefficient β is normalized to the unit drive and is not a carrier amplitude (SM Section 5.4).

5.2 Dependence on the nuisance span

Imported from prior work. For the causal (lagging) template, the registered-grid maximum of $|\hat{\beta}|/\sigma_F$ is 2.2851, with global $p=0.26$ under the recorded Gaussian-noise simulations, and the stored detection flag is false. Table 1 reports the envelopes for the normalized coefficient β . Neither $K=10$ column is a calibrated astrophysical 95 percent exclusion, and the truncated $K=1$ column, the narrowest stored construction, fails under omitted means (Section 5.3). Both $K=1$ columns use the stored diagonal weighting, which also fails under correlated extra Fourier power (Table 2); the full-space $K=1$ values come from the registered nuisance audit.

Table 1: Imported from prior work. Registered-grid envelopes U for the unit-drive coefficient β . Full and truncated use the stored 90- and 71-direction nuisance spaces. K is the width multiplier in Equation (16). The last column is the full/truncated ratio at $K=10$; the Full, $K=1$ column is from the registered nuisance audit.

τ_χ (day)	Full, $K=10$	Full, $K=1$	Trunc., $K=10$	Trunc., $K=1$	Ratio
2	3.534×10^{-9}	6.539×10^{-10}	1.680×10^{-9}	2.794×10^{-10}	2.10
5	9.673×10^{-9}	1.530×10^{-9}	1.852×10^{-9}	2.884×10^{-10}	5.22
18	3.166×10^{-8}	5.264×10^{-9}	1.976×10^{-9}	2.933×10^{-10}	16.02
52	4.488×10^{-8}	9.526×10^{-9}	2.600×10^{-9}	3.669×10^{-10}	17.26
200	1.241×10^{-7}	2.730×10^{-8}	7.155×10^{-9}	9.955×10^{-10}	17.35

Imported from prior work. Over the 65 stored lags the full/truncated ratio at $K=10$ ranges from 2.10 to 17.41 (Table 1 shows five lags), and at $K=1$ from 2.34 to 27.5 over the six audited lags. This dependence is part of the result, not a small correction to the narrowest interval. An SVD audit finds numerical rank 90 with minimum relative singular value 4.18×10^{-7} ; it does not certify the finite-difference derivatives. The weak modes mix timing, astrometric, planet and Fourier directions, and no physical prior excluding that subspace has been established. Full rank is therefore the primary baseline below. Figure 1 displays the comparison. **Conjectural.** Because the weak directions rotate strongly under recomputed derivatives (Section 5.6), part of the full/truncated difference may reflect numerical derivative error.

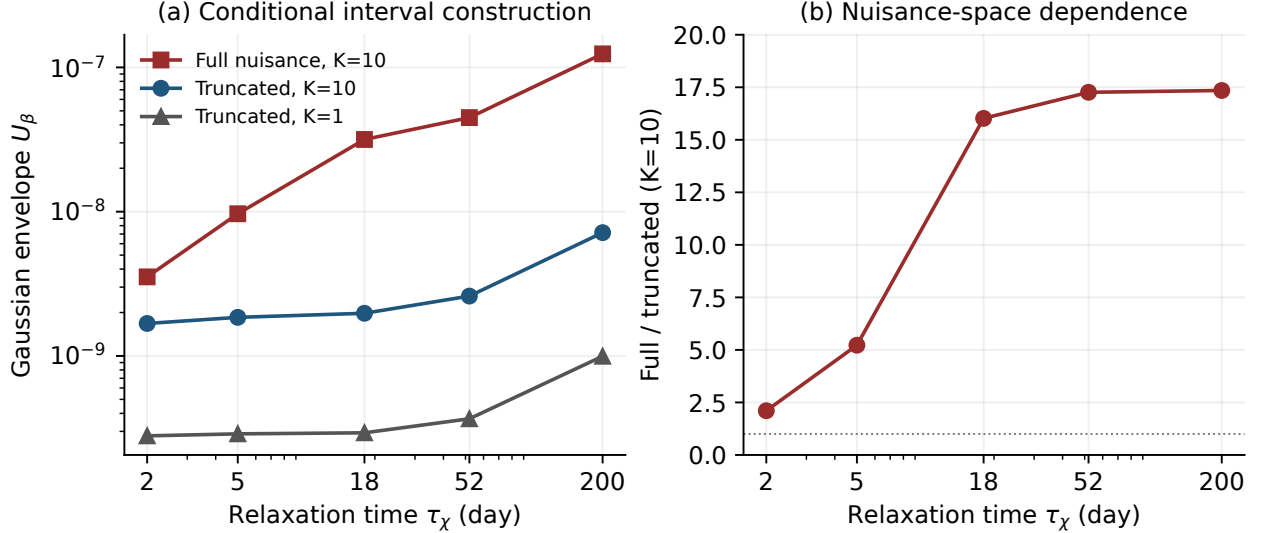


Figure 1: Imported from prior work. Stored interval envelopes and nuisance-space dependence. Left: $K=1$ and $K=10$ truncated constructions and the $K=10$ full-space scenario. Right: the full/truncated $K=10$ ratio. Connecting lines guide the eye; they do not certify unsampled lag or phase values. These are normalized-coefficient intervals, not universal SEP exclusions.

5.3 Coverage of the interval construction

Imported from prior work. A registered experiment uses 8192 independent realizations per condition, 18 preselected lag/origin cells and nine amplitudes ($0, \pm 2, \pm 5, \pm 20$ and ± 50 full-space unit-noise σ). The stresses are omitted means of norm 3, 10 or 30 and extra Fourier modes 31–60, with spectral variance proportional to j^{-4} normalized to weighted RMS 0.25 or 1. Table 2 reports pointwise coverage minima.

Imported from prior work. Truncation fails under omitted means, and diagonal weighting fails under correlated extra Fourier power; the oracle GLS control, which uses the generating covariance, restores approximately nominal coverage. A follow-through registered after those outcomes estimates $C = \sigma^2(I + a^2 LL^T)$ by restricted maximum likelihood (REML) [18], a simple form of the correlated-noise likelihoods used in pulsar timing [19], where the columns of L are the extra Fourier modes and a is their amplitude, the same fixed family that generates the stress; across 486 conditions the minimum $K=1$ coverage is $7749/8192=0.945923$. A truncated pointwise $K=10$ interval also has a $0/8192$ stress case, while separately registered complete-grid tests at lags 2 and 200 days give $K=10$ coverage of at least $511/512$ in every tested scenario. These are Monte Carlo

Table 2: Imported from prior work. Minimum pointwise coverage of the K=1 U construction across tested cases. Each condition has 8192 realizations. Minima are descriptive, not simultaneous guarantees. Estimated covariance is a separately registered follow-through; dashes denote conditions not rerun for that estimator.

Generating condition	Trunc. diag.	Full diag.	Oracle GLS	Est. GLS
White noise	0.9468	0.9481	0.9481	0.9473
Omitted mean, norm 3	0.0956	0.9481	0.9481	–
Omitted mean, norm 10	0	0.9481	0.9481	–
Omitted mean, norm 30	0	0.9481	0.9481	–
Extra Fourier RMS 0.25	0.9310	0.7469	0.9485	0.9471
Extra Fourier RMS 1	0.8132	0.5825	0.9490	0.9459

results within the prescribed covariance families, not an independently validated astrophysical noise model.

5.4 Information left after derivative and fast-spectrum comparators

Imported from prior work. After full nuisance removal and covariance whitening at the fixed amplitudes $a = 0, 0.0831611, 1$ (the middle value from local REML fits to the stored residuals), the six-carrier response maps have rank six and condition numbers 98.19–124.82. The retained information of a comparator is the Fisher information for β after co-fitting the comparator’s columns, relative to co-fitting only the instantaneous column. Its minimum over six lags and three specified phase sets (zero phases, one fixed independent set, and the leading physical phases of Section 4, all with unit carrier amplitudes) is 0.005389 at N=1, 6.29×10^{-5} at N=2, 1.17×10^{-5} at N=3, and 4.63×10^{-6} at N=4; the last widens the unit-noise standard error by about 465 times. The even-only comparator $\{0, 2, 4\}$ retains at least 0.0311. Mathematical noninterpolation can therefore leave very little measurable information. **Proven.** At N=5 the information is exactly zero by Theorem 1; numerical residuals between 5.4×10^{-30} and 5.3×10^{-25} are rounding error around that identity. **Imported from prior work.** Projected through the same maps, the inequality of Equation (6) gives a positive distance witness for every tested lag with the explicitly illustrative $\Lambda = 10\omega_{\text{in}}$ (SM Section 5.6). This is a conditional discrimination calculation, not a measured gap.

5.5 Physical drive and simultaneous inference

Imported from prior work. With the runtime’s parameter set, the leading physical drive of Equation (13) has $U_* = 1.74999156 \times 10^{-10}$, normalized amplitudes (0.24394925, 0.69195682, 0.06409393), and the phases implied by the pericenters of Section 4. Exact coplanar Kepler-potential grids give an omitted-input RMS of 3.37427 percent of the leading-drive RMS; this is not an omitted timing-error bound. **Proven.** Fitting all six carrier coefficients θ before a lag or phase is selected lets a confidence region E for them be inverted through the drive map $\theta = W(\phi, \tau)b$, $b = (c_Y, \beta)$: the physical lag section at lag τ is the set of b with $W(\phi, \tau)b \in E$, and this controls continuous phase/lag selection within the declared mean model. **Imported from prior work.** A six-coefficient REML region calibrates a quadratic threshold of 12.8241766 from 8192 draws at each $a = 0, 0.25, 1, 4$, frozen before independent validation, whose inclusion rates are 0.951660, 0.954346, 0.952759 and 0.954102; these support the specified procedure, not every astrophysical noise process or intermediate covariance. The recorded data’s six-coefficient statistic, 16.3525, lies above that threshold (nominal chi-square

p of about 0.012 for six degrees of freedom), so the data reject vanishing carrier coefficients. This test differs from the unit-drive scan of Section 5.2 ($p=0.26$) in statistic, template and covariance: it tests all six carrier coefficients at once. The best fit within the physical plane leaves a statistic of 12.84 at 2 days and 14.39–14.59 at the other five evaluated lags. Every evaluated physical lag section is therefore empty: at these six lags, no instantaneous term plus single relaxation driven by the leading physical drive reproduces the carrier coefficients at the calibrated 95 percent level, and no interval on β or EOS-matched bound follows. The 2-day rejection is marginal: the frozen threshold is the largest of four calibration order statistics (12.27–12.82), each with a Monte Carlo standard error near 0.13, although 12.84 exceeds all four and the nominal chi-square value 12.59. **Proven.** The rejection is robust for the equal-charge realization of Section 4, which has $\beta \geq 0$. The statistic is a convex quadratic in b , the $\beta = 0$ line lies in every section, and the unconstrained $\hat{\beta}$ is negative at five lags and 7.4×10^{-12} at 18 days; so with $\beta \geq 0$ every section minimum is at least the 18-day value, 14.59. **Conjectural.** These results use the archived timing derivatives, which Section 5.6 shows to be uncertain. At the evaluated lags the excess is not explained by a response to this drive, so it gives no evidence for relaxation; most of it lies in directions that the drive does not span, and it is not attributed. Timing-model mismatch at the orbital carriers, the derivative errors of Section 5.6, the omitted drive terms and covariance misspecification are candidates that this analysis does not separate.

5.6 Limits of the stored record

Imported from prior work. At $\tau_\chi = 500$ days, one-per-mille settling of a transient requires 3453.88 days, longer than the stored observing span. On the stored sampling, 0.7984–0.9973 of an exponential input’s norm lies outside the constant-plus-six-periodic input span across the tested lags; this is input independence, not a timing response. Dedicated live transient columns at $\tau = 2, 52, 500$ days pass a registered 5-percent convergence gate. Co-fitting the measured transient changes the physical-drive β standard errors by factors 0.999998–1.00742 across the nine tested lag/covariance combinations; this does not certify every lag, large initial amplitudes, or coverage after adding that coefficient.

Imported from prior work. Recomputing the 28 timing derivatives at half the archived steps leaves both nuisance matrices at rank 90 but gives a maximum principal sine of 0.999913, changes the physical-drive standard errors by factors 0.7858–0.8722 and shifts the estimates, so a rigorous derivative-error budget remains unavailable. Among the 201 pulse-count assignments with at most two nonzero ± 1 turn steps on the ten longest gaps, the weakest is a single step at a 223.37-day gap, with minimum chi-square changes 27.9406, 11.6183 and 0.152887 at $a = 0, 0.0831611, 1$. Its unconstrained linear compensation implies eccentricities above one and is rejected as a physical fit; this does not exclude a different nonlinear solution. SM Sections 5.5–5.10 record these audits in full.

6 Discussion

Proven. Finite-order static representation and dynamical identifiability are different questions. A settled single pole escapes a restricted derivative comparator, but finitely many carriers are interpolated once enough coefficients are admitted (Theorem 1), and nuisance freedom can erase the remaining functional distinction. Positivity with a rate gap restores a comparator restriction, and two calibrated samples can identify one observable pole, but not the number of internal variables behind it.

Counterexample candidate. The physical target is a shared transfer relation with known drive and constrained projection, tested against an explicitly bounded comparator. The one-state model is one realization; hereditary kernels, multiple states, nonlinear thresholds and other sectors remain separate possibilities.

Imported from prior work. For our stored J0337 analysis, the interval on the relaxation coefficient depends strongly on the retained nuisance span, and calibrated inference is available within specified covariance families. At six evaluated lags between 2 and 500 days, the recorded carrier coefficients carry an excess that the leading physical drive, with or without a single relaxation, does not reproduce; no relaxation is detected. The physical-drive fits have standard errors of 3.5×10^{-10} to 1.7×10^{-9} in β , about 2 to 10 in $B = \beta/U_*$. The measured derivative sensitivity and the unphysical full pulse compensation keep the analysis from serving as an empirical constraint.

Conjectural. SM Section 4.6 works through one internal state of the inner white dwarf of the triple, the scalar-driven displacement of its own matter, in a specified weak-field scalar-tensor model. A scalar pulse leaves in the displaced matter a transient retarded monopole signal. Under the stated assumptions, which include linear response, $|a_p| \leq 1$ and an outer-companion charge near the Cassini limit, its structural modulation of the pair factors at orbital timescales lies about eight orders of magnitude below the smallest stored Section 5 scale. A prescribed layer-by-layer relaxation estimate, not a non-adiabatic calculation, gives contributions from the scanned layers of 4.0×10^{-9} and 3.3×10^{-7} of the structural term at the inner and outer orbital frequencies. The scan stops at a thermal time of 10^{13} s; deeper layers have an uncomputed relaxation strength, so these partial sums do not bound the total lag without the tail condition stated in SM Section 4.6. These responses enter the pulsar-inner and inner-outer pair factors, not the common modulation of Section 4. The white dwarfs’ instantaneous zero-lag responses could reach the smallest stored scale only for $|a_p| \gtrsim 0.5$, which the static SEP limits allow only if $|a_o| \lesssim 5 \times 10^{-6}$; they are not lags. Neither timing response was computed, so this is a scale comparison, not an exclusion. Other internal states of the white dwarfs, and the neutron-star charge of Section 4, remain open.

Conjectural. A physically calibrated exclusion still requires numerical EOS-to-body matching, control of omitted drive/force terms, a reliable derivative space and an adequate astrophysical likelihood. The fast-rate gap remains an unestablished physical premise, and no mechanism that produces a relaxation at the stored 2–500-day lags with a coupling B of that size is identified here. The result is a set of identifiability boundaries and a conditional application, not a universal empirical SEP bound.

Use of AI tools

This work used AI agents substantively, under the author’s direction and responsibility. Coding agents built on Claude Opus 5.5 (Anthropic, run through Claude Code) and GPT-6-Astra (OpenAI, run through the Codex command-line tool) wrote and ran most of the analysis, stellar-structure and verification code, carried out the computations reported here, and drafted and revised manuscript text, including scientific claims and explanations. GPT-6-Astra, Claude Opus 5.5 and Claude Fable 5.1 (Anthropic, run as a Claude Code subagent) independently reviewed drafts of the white-dwarf calculation (SM Section 4.6) in several rounds and the complete manuscript before submission; their reports and the responses are archived in the repository. The author set the research questions, the acceptance criteria and the claim labels, and checked the AI output against the stored numerical records, the SHA-256-bound manifests, symbolic checks, the verification program named below and these reviews.

Statements and declarations

Competing interests

The author declares no competing interests.

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Author contributions

J.K. conceived and directed the study, set the research questions and acceptance criteria, checked the results against archived numerical records and verification tests, and reviewed and approved the manuscript. AI assistance with coding, computation, drafting and internal review is disclosed in the Use of AI tools section. J.K. takes responsibility for the work.

Data and code availability

The public timing release is cited as a dataset [10]. The manuscript and SM sources, code, notes, manifests and numerical results are available in the public snapshot tagged `grg-submission-2026-09-28` at <https://github.com/lpaiu-cs/self-mass-unobservability>. The accompanying `paper/revision-manifest.json` identifies this revision's inputs by SHA-256, and `output/submission/grg/manifest.json` binds the submission files. Binary runtime arrays, about 50 GB in total, exceed the hosting limits and are not deposited; the manifests identify them by SHA-256, and they are available from the author on request. Commit identifiers cited in the SM refer to the author's full repository history, of which the public repository holds a snapshot; `PUBLIC_SNAPSHOT.md` there states what the snapshot omits.

Online Resource 1 (`ESM_1.pdf`): complete technical account, static operator catalog, derivations, numerical audits and white-dwarf calculation. The SM source is `output/submission/grg/supplement.md`. It contains the static operator catalog, all derivations, the full audits with their registration commits and reproduction commands, and the white-dwarf calculation with a map from each result to its manifest. The public JSON record `outputs/research-completion/public-inference.json` supplies the six fitted coefficients, their precision matrix, the physical-drive map and the frozen threshold. With NumPy, `verification/replay_public_inference.py` reproduces the omnibus statistic and the six physical lag sections from this record without private arrays. It does not validate the timing derivatives or recalibrate coverage. The broader `verification/verify_unified_paper.py` also checks finite-carrier interpolation, stored audits, Tables 1 and 2, and input hashes; it requires the omitted arrays but does not invoke Numpy. Outputs superseded by the periastron-convention correction of Sections 4 and 5 are kept in `outputs/research-completion/withdrawn-periastron-convention/`.

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